

Build A Model Of Satellite Dish Antenna Angle Controller

Ashutosh M. Bhatt

Presented here is an Arduino-based dish antenna controller to align the position of a satellite dish antenna by setting azimuth and elevation angles. The dish antenna

should always point to a specific satellite in space to receive maximum signal. Even a slight deviation from the point may reduce signal strength.

Azimuth and elevation angles are set using input switches, as explained in the next section. Once these values are set, the dish antenna automatically moves in vertical and horizontal directions with the help of two DC geared motors.

Fig. 1 shows the diagram of the antenna system along with the positions of azimuth (AZ axis) and elevation motors (EL axis).

Circuit and working

The circuit diagram of the satellite dish antenna angle controller using Arduino is shown in Fig. 2. It is built around Arduino Uno (Board1), two slotted opto-interrupter sensors MOC7811 (IC1 and IC2), motor driver L293D (IC3), 20x4 liquid crystal display (LCD1),

two 12V DC, 10rpm motors (M1 and M2) and a few other components.

Each sensor circuit is built around MOC7811 having internal IR LED and phototransistor. Both sensor circuits for elevation and azimuth angles are identical. Internal IR LED is given 5V supply through 330-ohm current-limiting resistors (R6 and R8), and photo-transistor is connected to switch configuration using external 1-kilo-ohm resistors (R7 and R9).

IR light beams continuously fall on the phototransistors. Sensor outputs are taken from collectors (pin 2) of the phototransistors.

Both sensor outputs (from IC1 and IC2) are connected to digital I/O pins 2 and 3 of Arduino Uno, respectively. Pins A0 through A4 are connected to 5V through resistors R1 through R5, respectively. Switches S1 through S5 are connected such that when a switch is pressed, logic 0 is given as input to Arduino Uno.

Digital pins 8 and 9 of Arduino Uno are connected to LCD1 control pins RS and EN, respectively. LCD1 data pins DB4 through DB7 are connected to digital pins 10, 11, 12 and 13 of Arduino Uno. R/W pin of LCD1 is connected to ground to make it write enable. Digital pins 4, 5, 6 and 7 of Arduino Uno are connected to input pins of L293D (IC3). The motors are connected to output pins of L293D. Arduino Uno drives the two DC motors (M1 and M2) through L293D.

The whole circuit works on 5V DC supply. Arduino Uno can be given 5V regulated supply on its 5V pin. A separate 12V supply is given to pin 8 of

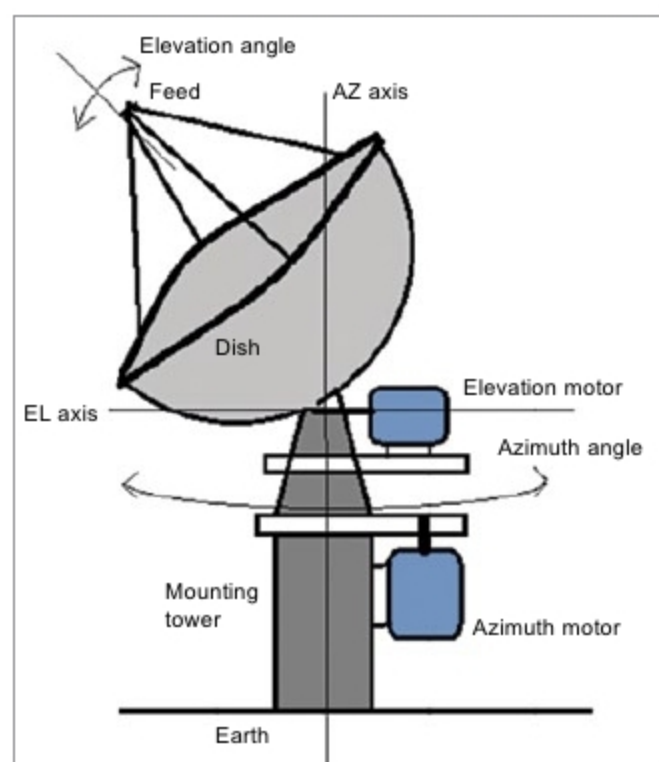


Fig. 1: Satellite dish antenna system

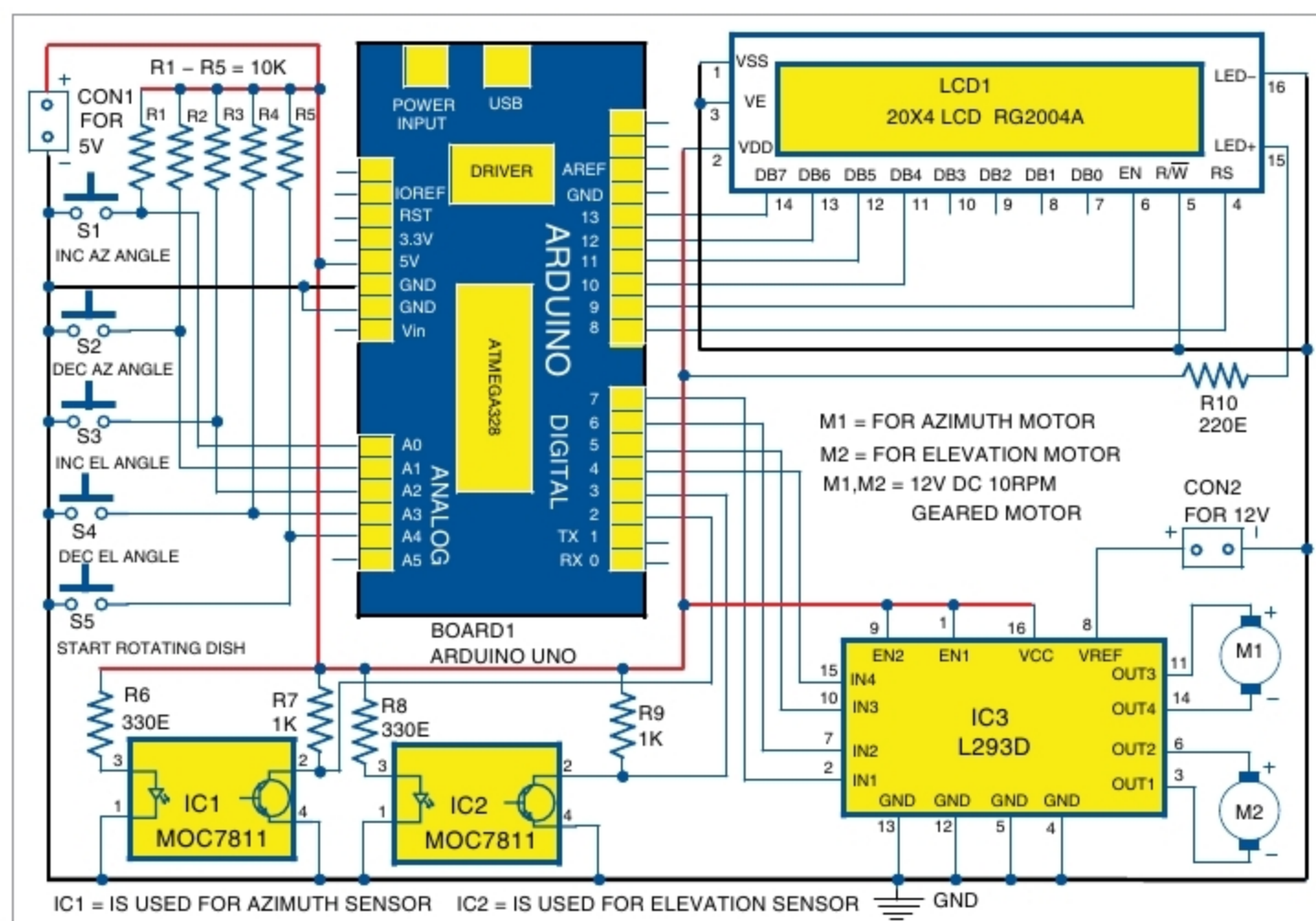


Fig. 2: Circuit diagram of the satellite dish antenna angle controller